Simulated Agent Populations with Diverse Reciprocity Pre-Dispositions to Model Real-Time Helping Behavior

Harrison Wolff [1] Divya Srinivasan [1] Nirvan Shukla [1]

Abstract

This study builds upon the experimental framework of Osborn Popp and Gureckis (2024) examining factors influencing moment-to-moment helping decisions in a resource-limited, video-game-like environment involving 314 participant pairs. Analyzing the dataset of 58,882 individual agent interactions collected by the experiment yields insights used to construct a novel, diverse population of nearest neighbor agents whose aggregate behavior across simulated games closely mimics the observed behavior of human players. These agents compute distinct probabilities of helping their partner in real-time, varying at initialization by the parameters governing their baseline pre-disposition to rendering helping actions, as well as the way they respond to the helping actions of their partner and other environmental variables. Analyzing the relative performance achieved (i.e., score) in the game environment as a function of agent parameter initialization also yields insight into why the significant diversity of reciprocal behaviors exhibited by both agents and humans may be present.

1. Introduction

In our interconnected world, understanding the dynamics of cooperative behavior is crucial, especially when individuals must decide when and how to help others, often at an opportunity cost to themselves. Earlier this year, Osborn Popp and Gureckis published an empirical study called *Moment-to-Moment Decisions of When and How to Help Another Person* (Osborn Popp & Gureckis, 2024), which asked this very question. Their research explored several factors influencing ”helping behavior” in a multi-agent environment through an experimental setup which mimicked a farming video game.

Their study involved 314 pairs of human participants tasked with gathering vegetables in a resource-limited environment. Each participant was assigned a role as either a ”red farmer” or a ”purple farmer,” and had a finite amount of energy to expend by collecting vegetables to increase their own score or by collecting vegetables that solely benefited their partner, with the hope that their partner would later reciprocate their efforts. The authors found that helping behavior was more likely when the energy cost was low, the player’s energy was high relative to its starting level, and there was a reasonable expectation of reciprocity. The dataset includes $58,882$ turn-level observations from each of the 628 participants, capturing detailed information such as player roles, actions taken, spatial positioning, resource costs, and visibility conditions (whether players could see their partner’s score).

Leveraging empirical relationships between helping behavior, the helpfulness of a player’s partner, and environmental variables in the data collected by the experiment, we create a global model calculating the overall probability of helping behavior across the entire player population at the player/turn level. We then use our global model as a prototype for inferring a **distribution of individual model parameters** across human players which in aggregate equate to the global model, yielding a generating process for a diverse group of agents whose aggregate behavior can more closely model the aggregate behaviors of the human population than a single uniform model. A modified version of the nearest neighbor simulated agents developed by Popp and Gureckis is then used to construct a population of simulated agents, whose behavior is compared with that of the human population.

Exploring how humans might ”track” reciprocal exchanges over time, as discussed in much of the related literature we looked at, could offer some valuable insights. Boyd’s article *Is the Repeated Prisoner’s Dilemma a Good Model of Reciprocal Altruism?* evaluates the prisoner’s dilemma as a model for reciprocal altruism, arguing that its assumptions about memory and interaction patterns do not translate to the real world. By proposing a mathematical model that incorporates memory constraints, partner variability and the cost/benefit ratio of cooperation, Boyd’s work challenges traditional perspectives and underscores the need for taking into account a real life environment when modeling cooperation between humans (Boyd, 1988).

The study (Volstorf et al., 2011) builds upon prior research into memory strategies, focusing on mechanisms like the "cheater-memory" and "rarity" hypotheses. Existing work suggests that individuals may preferentially remember either cheaters or rare partner types as a way to reduce cognitive effort. Research has also highlighted the challenges posed by error-prone memory in tracking specific actions, pointing to the utility of stable partner categorizations, such as labeling individuals as cooperators or defectors. These categorizations reflect evolutionary strategies that adapt behavior based on recent interactions, emphasizing the role of reciprocity. This body of knowledge illustrates the interaction between memory limitations, ecological considerations, and strategic decision-making in fostering cooperative behavior. The present study builds on these foundations by incorporating repeated interactions in a simulated environment, testing memory retention over both short and long intervals to evaluate the robustness of these strategies in dynamic settings.

2. Methods/Models

Using the raw turn-level data and features collected by (Osborn Popp & Gureckis, 2024) ($n = 58882$), we seek to:

1. Identify the most relevant features predicting when a player will help another player (i.e., pick their partner's color vegetable as opposed to their own.) This will include features derived from both past helping actions as well as the game environment on a given turn.
2. Build a global, population-wide model explaining the overall probability of rendering help for all 628 players in the available population, conditioned on our relevant features.
3. Extend this model to a distribution of individual-level models, explained by a generating distribution of parameters at the individual level which "average out" to our global model, explaining differences in individual behavior that cannot be explained by our predictor variables alone.

2.1. Feature Selection

To facilitate feature selection, we first filter out all datapoints corresponding to end-of-game ($n = 3768$), trips to the farmbox to deposit collected vegetables ($n = 11874$, which aren't relevant to our analysis), and turns where no action is taken ($n = 5500$, which correspond to a rare action that won't be considered as part of our cognitive model.) This leaves $n = 37680$ individual player interactions for our analysis. We then filter for all numerical features with an empirical absolute correlation of at least 0.1 with whether or not a player helped on a given turn (the "helped" feature), yielding the corelation matrix shown in Figure 1. ("helped" corresponds to the bottom-most row and right-most column.)

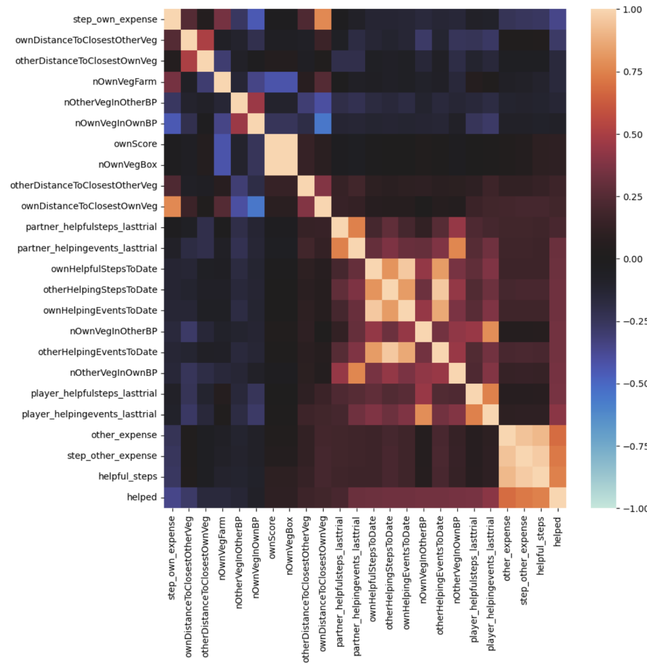

Figure 1. Correlation matrix of all features with absolute correlation ≥ 0.1 to "helped"

Features with a tautological correlation to "helped," such as "helpful_steps" and "step_other_expense" and "other_expense" which show a near 100% correlation to it, are discarded. "otherHelpingEventsToDate" is then taken as a proxy for the large mutually correlated block of features of which it is a member, spanning from "partner_helpingevents_lasttrial" to "player_helpingevents_lasttrial" in Figure 1, all of which are either too correlated to "otherHelpingEventsToDate" to add meaningful additional information, or are again tautologically correlated to "helped."

A similar reasoning process on all features in Figure 1 to the left of "partner_helpfulsteps_lasttrial" leaves us with only 5 relevant features around which to build a model, which are shown in Figure 2. Given their lower correlations to "helped" and lack of significant value add during modeling, "otherDistanceToClosestOwnVeg" and "otherDistanceToClosestOtherVeg" are further removed. This leaves us with only 3 necessary features for cognitive modeling: "otherHelpingEventsToDate", "ownDistanceToClosestOwnVeg", and "ownDistanceToClosestOtherVeg."

This analysis suggests that the environmental variables affecting a player's probability of helping their

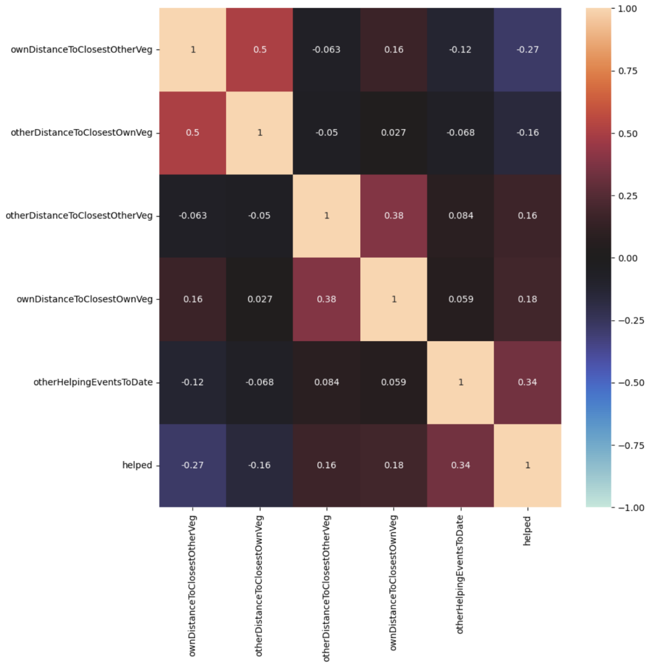

Figure 2. Correlation matrix of pruned features from the original set, which numerical correlations displayed

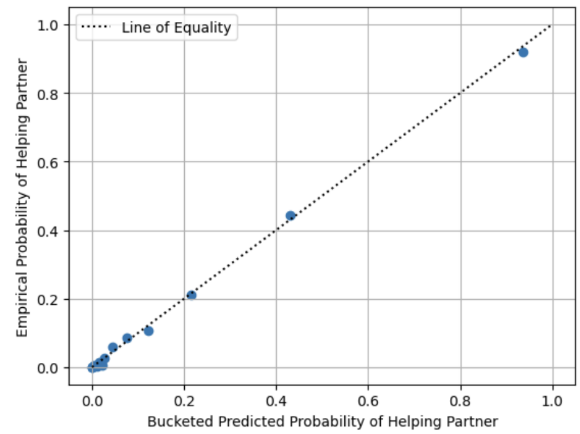

Figure 3. Bucketed Out-of-Sample Predicted Helping Probabilities vs Empirically Observed Helping Probabilities

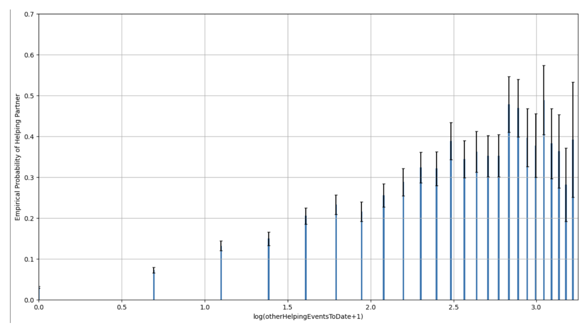

Figure 4. Overall Empirical Probability of A Person Helping Their Partner vs $log(otherHelpingEventsToDate + 1)$. Errors bars correspond to 95% confidence intervals

partner, (not yet including inherent pre-dispositions to reciprocity for each player, which will be addressed later), can largely be distilled to how many times a player's partner has helped them across all games played thus far, how close they are to their closest own vegetable, and how close they are to their partner's closest vegetable. The first of these features provides an intuitive way to track the "overall helpfulness" of an agent's partner similar to the studies in our introduction, while relative distances measure relative cost of resource exploitation between helping and not helping at any given time.

2.2. Global Modeling and Feature Engineering

To evaluate the effectiveness of our selection of features, as well as the necessary model complexity to model their relationship with a player's probability of helping their partner on any given turn, we perform a global logistic regression over the entire dataset between our 3 remaining features and whether or not a player helped their partner on any given turn (encoded as "1" if they did and "0" otherwise.) $n = 25000$ turn-level interactions are selected as a training dataset, with the other $n = 12680$ used for out-of-sample evaluation. To improve out-of-sample model performance, we add 1 to and then take the natural log of "otherHelpingEventsToDate," as well as divide both "distance to closest veg" features by 5 and subtract 1 in order to "quasi-normalize" them to a mean close to 0 and a standard deviation close to 1. We also introduce L1 regularization term with $\alpha = 25$ to improve out-of-sample performance.

Table 1. Results of logistic regression global helping probability model. Errors shown correspond to 95% confidence intervals. Learned feature coefficients are shown in standard text, out-of-sample performance metrics are shown in bold.

Feature/Performance Metric	Value
log(otherHelpingEventsToDate+1)	1.203± 0.113
ownDistanceToClossetOwnVeg/5 - 1	1.323± 0.085
ownDistanceToClossetOtherVeg/5 - 1	-1.567± 0.082
const	-3.657± 0.114
Out-of-Sample Log Loss	**0.168**
Out-of-Sample AUC	**0.946**

Logistic regression results are shown in Figure 3 and Table 1, respectively, demonstrating very high predictive

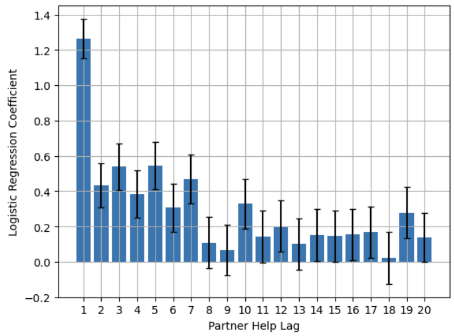

Figure 5. Logstic Regression Coefficient of the nth lagged indicator variable of an agent's partner helping them. Error bars correspond to 95% CI's

power from our selected features and little structural bias in the selected model. The coefficient values learned also make significant intuitive sense and corroborate the results of (Osborn Popp & Gureckis, 2024), with a player's probability of helping their partner increasing substantially if their partner has helped them frequently in the past or they are far from their own closest vegetable, and decreasing substantially if they are very far from their partner's closest vegetable. The significantly positive coefficient on $\log(\text{otherHelpingEventsToDate} + 1)$ also corroborates the significant increase in empirical helping probability as a function of how many times someone's partner has helped across the entire population, as shown in Figure 4.

In order to deeper understand the relationship between an agent partner's helping history and an agent's probability of helping, a second logistic regression was conducted with "otherHelpingEventsToDate" decomposed into independent features of whether an agent's partner helped on the last turn, the turn preceding the last term, and so on (more generally, the nth lagged term with respect to the agent's current turn, encoded as "1" if an agent's partner helped and "0" otherwise.) Quasi-normalized ownDistanceToClosestOwnVeg and "ownDistanceToClosestOtherVeg" are still included to control for any confounding relationship with an agent's partner's helping history, and the the coefficients found for each lagged partner helping term is shown in Figure 4. This analysis shows that recent helping actions from an agent's partner carry significantly more influence over an agent's decision to help their partner than those in the more distant past, suggesting that reciprocity is mostly a function of short-term memory of the most recent helping (or non-helping) actions observed from one's partner.

Further experimentation revealed that an exponentially weighted cumulative sum with a halflife of 4 turns is a simple and highly effective way to capture this dynamic, with more recent helping interactions from an agent's partner carrying a higher weight in the sum. This gives rise to the final features which we use to construct our agent-level co-operability models:

- f_1 = Expoentially-weighted cumulative sum, with a halflife of 4 turns, of an agent's partner's helping history across all games played thus far, encoded as a "1" if the partner helped on a given turn and "0" if they did not.

- $f_2 = \frac{ownDistanceToClosestOwnVeg}{5} - 1$

- $f_3 = \frac{ownDistanceToClosestOtherVeg}{5} - 1$

2.3. Distribution of Model Parameters Modeling Individual Reciprocity Dispositions

While our 3-variable logistic regression model performs very well explaining broad trends in helping probabilities across all players, the variation in helping behavior between agents and games shown in Figure 4 imply that inherent pre-dispositions to reciprocity (i.e., the parmaeters of our model), likely very widely among individuals. Rather than provide a universal model that can be applied to any individual, our global model is rather a template upon which a distribution of indivdiual level agent models can be derived. To deduce the distribution of model parameters amongst individuals which "average out" to our global model, all 628 particiants in our dataset are bucketed in 8 equally-sized quantiles based on the overall proportion of times they helped their partner, with "Quantile 1" containing individuals that helped the lowest proportion of times, and "Quantile 8" the highest. Each of these quantiles is then regressed indivdiually, from which a distribution of parameters characterizing indivdiual variation in our population can be bootstrapped. The results of these regression and correlations between model parameters across quantiles is shown in Figures 6 and 7.

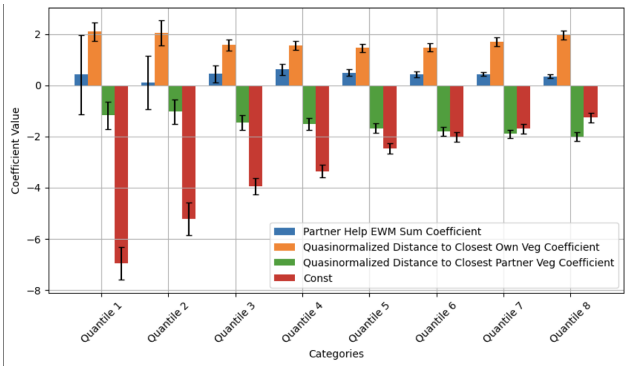

Figure 6. Learned Logistic Regression Coefficients by Quantile, with individuals bucketed based on the total proportion of times they helped their partner across all games played. Error bars correspond to 95% CI's

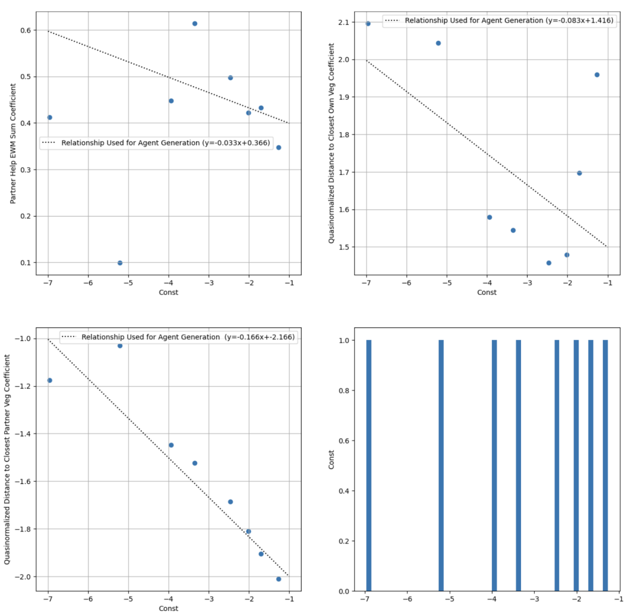

Figure 7. Scatter plots of quantile-level coefficients vs the constant term learned for each quantile. (Top left, top right, and bottom left.) As well as the distribution of constant terms across quantiles (bottom right)

These results illustrate that all individuals in our dataset exhibit reciprocity and distance sensitivity in the same direction, (i.e., all quantiles have a positive "Partner Help EWM sum" and "closest distance to own vegetable" coefficient, and negative "distance to closest other vegetable" coefficient.) The primary source of variation between individuals stems from variations in the constant term of their individual regression models, which represent a "baseline level" of helpfulness, which then increase or decrease by similar dynamics across the population. In simpler terms, some people start out more disposed to helping their partner than others, but all individuals' helpfulness will increase in response to their partner offering help, and decrease in response to being far from their partner's closest vegetable or close to their own closest vegetable. The distribution of constant terms across individuals is modeled as a mixture of uniform distributions, which in conjunction with the hand-fitted linear fits presented in Figure 7 give rise to the agent generating process described in Algorithm 1. w_1,w_2,w_3, and b are then used to describe the probability of an agent helping based on the values of f_1, f_2, and f_3 experienced by the agent at a given time.

Algorithm 1 Agent Generation Process Returning (w_1, w_2, w_3, b)

$p \leftarrow \mathcal{U}(0, 1)$

if $p \leq 0.5$ **then**
  $b \leftarrow \mathcal{U}(-2.5, -1)$
else
  $b \leftarrow \mathcal{U}(-10, -4)$
end if

$w_1 \leftarrow -0.033b + 0.366$
$w_2 \leftarrow -0.083b + 1.416$
$w_3 \leftarrow -0.166b - 2.166$
Return w_1, w_2, w_3, b

3. Results

To test our algorithm against human data, a synthetic population of 10,000 agent pairs with varying baseline levels of helpfulness and feature weights are generated using Algorithm 1. Each pair is then made to play 12 instances of the farm game using the same randomly generated resource condition, cost condition, visibility condition, and layout, each selected with uniform probabilities across all possibilities. The agent's actions, including whether they help their partner at any given step, are determined by Algorithm 2, with (w_1, w_2, w_3, b) used to determine the probability an agent helps their partner at each turn based on their partner's action and their distances to their own and their partner's closest vegetables.

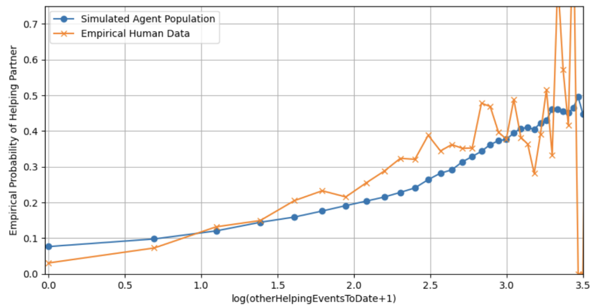

Figure 8. Simulated Agent Population-Wide Empirical Helping Probability vs $log(otherHelpingEventsToDate + 1)$, compared with empirical human data

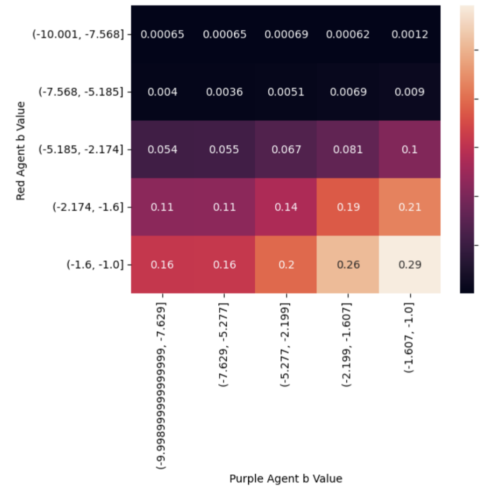

Figure 9. "Red" agent helping probability across all interactions as a function of its bias term and the bias term, (grouped as 5 equally sized quantile buckets across the agent population), of its partner. "Forced" helping events where an agent just doesn't have any of its own vegetable left on the borad are excluded, as are trips to the box.

Algorithm 2 Agent Behavior On a Given Turn, Based on Their (w_1, w_2, w_3, b)

Update f_1 based on whether or not the agent's partner helped them on the last turn. If their partner went to the box on their last turn, don't count this towards f_1

Compute f_2 and f_3 as described in section 2.2
Compute $p_{help} = \frac{1}{1+\exp(-(w_1 f_1 + w_2 f_2 + w_3 f_3 + b))}$

Generate a random variable $x \sim \mathcal{U}(0, 1)$

if $x \leq p_{help}$ **then**
  Pick the closest of the **agent's partner's color vegetable**, or go to the box if their backpack is full
else
  Pick the closest of the **agent's own vegetable**, or go to the box if their backpack is full
end if

Empirical probabilities of all agents within the population vs $\log((otherHelpingEventsToDate) + 1)$, as derived from their partner's actions with an identical methodology to the original dataset, is then compared to that observed in the empirical human data. Helping probabilities as a function of agent and partner parameters (excluding instances where an agent has none of their own vegetables left on the board and is "forced" to help their partner), as well as the average scores achieved by different types of agents, are also computed. To facilitate this, agents are bucketed based on the b term, (i.e., the constant term from the regression models in the previous section)

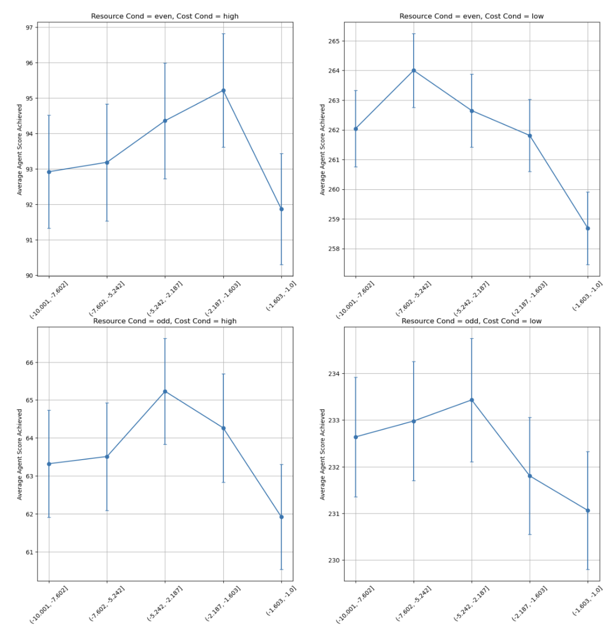

Figure 10. Average agent-level score across all games, bucketing by their bias term bucket (x-axis) and environment. Error bars denote 95% confidence intervals.

4. Discussion and Conclusion

As shown in Figure 8, our agent population's global helping probabilities as a function of their partner's cumulative helping actions to date closest mimics that observed in the human data collected by Popp and

Gureckis. It must be noted, however, that our agent distribution shows somewhat more frequent helping probabilities in cases when their partner has rendered no help ($\log(\text{otherHelpingEventsToDate} + 1) \approx 0$), which, in order to best fit our generating distribution to human data, requires we somewhat underestimate helping probabilities in cases where one's partner helps very frequently. One possible explanation is the existence of a special "retaliatory mechanism" not included in our model, reserved for cases when an agent's partner never co-operates, even when an agent renders them help. Our results in Figure 9 show that this behavior is observed among a significant minority ($\sim 30\%$) of the agent population, highlighting its importance in making optimal helping decisions.

Elaborating further on the results in Figure 9, we observe that the qualitative behavior of our agent population can be broken out into two primary groups:

1. Agents that will almost never provide help at the outset of a game sequence, and won't help their partner even if their partner is has a highly co-operative predisposition. (Top 2 rows, $\sim 30\%$ of the agent population)

2. "Tit for tat" agents ($\sim 70\%$ of the agent population), that are predisposed to varying degrees of helpfulness (baseline helping probabilities between 5 and 20%, as evidenced by their behavior when paired with uncooperative agents), that then become more helpful if helped by their partner. (An approximately 2x increase in probability of helping if paired with another highly co-operative agent, bottom 3 rows.)

Average game-level scores by agent bias bucket, as shown in Figure 10, provide a possible explanation for why this large diversity of agent pre-dispositions exist. While subtle, agents with less co-operative pre-dispositions (i.e., more highly negative bias term), tend to fair better in low cost environments by a statistically significant margin, while more co-operative agents tend to fair better in high cost environments. This is particularly prevalent in the "even" resource condition games, whose results are shown in the topmost row of plots. Evolutionarily, this dichotomy could serve as a proxy for rapidly alternating times of abundance and scarcity respectively, with the diversity of the overall group well-adapted to do well in both conditions.

In conclusion, this study provides a comprehensive exploration of cooperative behavior in a multi-agent environment, combining empirical data analysis with computational modeling to better understand the dynamics of helping decisions. By identifying key predictors such as partner helping history and spatial proximity (i.e., cost of exploiting) resources, we developed a logistic regression model that accurately predicts helping behavior in the global human populations. Furthermore, the incorporation of individualized agent parameters generated by our generating algorithm successfully captures the diversity in human pre-dispositions to co-operative behavior.

References

Boyd, R. Is the repeated prisoner's dilemma a good model of reciprocal altruism? *Ethology and Sociobiology*, 9(2-4): 211–222, 1988. doi: 10.1016/0162-3095(88)90029-3.

Osborn Popp, P. J. and Gureckis, T. M. Moment-to-moment decisions of when and how to help another person. *Proceedings of the 46th Annual Conference of the Cognitive Science Society*, 2024.

Volstorf, J., Rieskamp, J., and J.R., S. The good, the bad, and the rare: memory for partners in social interactions. *PLoS One*, 2011. doi: https://doi.org/10.1371/journal.pone.0018945.